

Answer **all** questions in the spaces provided.

1. Ash dieback disease is a fungal infection that is common in ash trees in Wales.

The disease causes the leaves to blacken and wilt, so eventually the ash tree has no leaves.

The photographs show healthy ash leaves and those affected by ash dieback.



- (a) Photosynthesis occurs in green leaves.

- (i) Use word(s) from the box to complete the paragraph below. You may use each word once, more than once or not at all. [5]

stomata	nitrogen	chlorophyll	oxygen	glucose
	water	carbon dioxide		

Leaves use to absorb light energy.

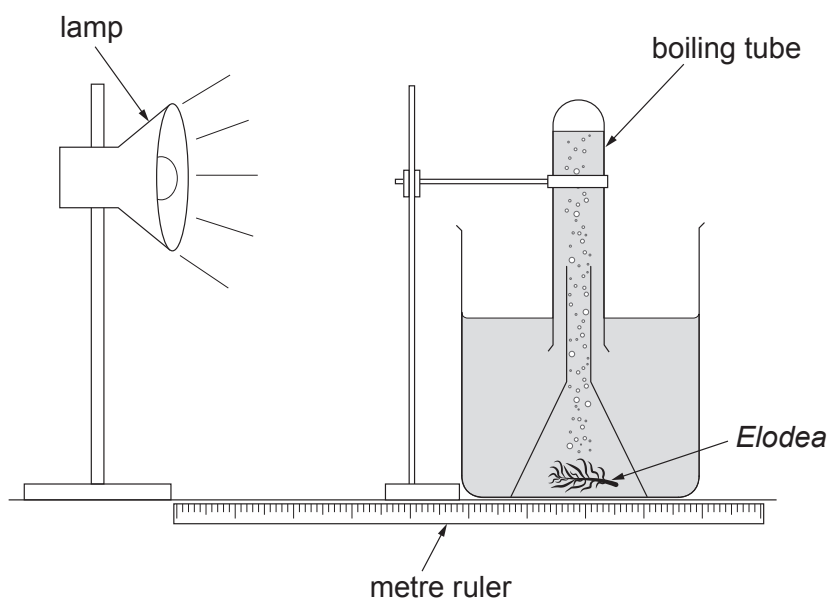
They convert and into
..... and

- (ii) Complete the following sentence by underlining the correct word in the brackets. [1]

Ash dieback disease affects the growth of an ash tree as the leaves are unable to make (**food** / **water** / **nitrogen**).



- (b) The following apparatus is used to investigate how light intensity affects the rate of photosynthesis of an *Elodea* plant.



The method used is described below.

1. Add **one spatula** of sodium hydrogen carbonate to a beaker containing 200 cm³ of water at 20°C, to supply carbon dioxide.
2. Place the *Elodea* in the beaker.
3. Completely fill a boiling tube with water and carefully place over the end of the funnel with the end under the water, clamp into place.
4. Place the lamp 5 cm away from the *Elodea*.
5. Start the stopwatch and record the **number of bubbles** of oxygen produced in a **few seconds**.
6. Repeat the experiment with the lamp 10 cm, 15 cm, 20 cm, 25 cm and 30 cm from the apparatus.

- (i) Steps 1 and 5 could lead to uncertainties. In the table below, circle **one** improvement in each row that you would use.

[3]

Current method	Improvement	
add one spatula	add 1 kg	add 1 g
record the number of bubbles	collect the gas in a beaker	collect the gas in a measuring cylinder
produced in a few seconds	produced in 1 minute	produced in 1 hour

- (ii) State how you would change step 6 to investigate the effect of temperature on photosynthesis.

[1]

.....

.....

10



- (b) (i) State **two** advantages of the materials used in 2015 over those used previously. [2]

1.

2.

- (ii) State **one** disadvantage of the materials used in 2015 over those used previously. [1]

.....

- (c) GRP is a common composite material, in which glass fibres are mixed with epoxy resin.

- (i) A GRP panel contains a volume of 300 cm^3 of epoxy resin.

Use the equation: [3]

$$\text{mass} = \text{density} \times \text{volume}$$

to calculate the mass of the resin.

mass = g

- (ii) I. State the strength of glass fibre in N/cm^2 . [2]

$$(1 \text{ MPa} = 100 \text{ N/cm}^2)$$

strength = N/cm^2

- II. The cross-sectional area (csa) of a glass fibre is 0.00015 cm^2 .

Use your answer in part (c)(ii)I. and the equation:

$$\text{force} = \text{strength} \times \text{csa}$$

to calculate the force required to break it. [2]

force = N

END OF PAPER





THE PERIODIC TABLE

Group

1

2

3

4

5

6

7

0

1 H Hydrogen 1										4 He Helium 2									

Key

A_r	relative atomic mass
Symbol	
Name	
Z	atomic number